Solutions to Mock JEE Main - 17 |JEE-2022

PHYSICS

SECTION-1

1.(A) $F = \eta A dv / dx$

$$\eta = F \frac{dx}{dv}\frac{1}{A} = m \times a \times \text{time} \times \frac{1}{\text{area}} = \frac{MV}{A'}$$

2.(B) Applying conservation of momentum, $m(u\hat{i} + 2u\hat{j}) + 3m(0) = 4m(v\hat{i} + v'\hat{j})$

$\Rightarrow \quad v = u/4$ and $v' = u/2$

3.(C) $\frac{hc}{\lambda} = 13.6Z^2\left(\frac{1}{1} - \frac{1}{4}\right) = 13.6Z^2\left(\frac{3}{4}\right)$; $\quad \left[\lambda = \frac{4}{3} \times k\right]$

β line of Paschen

$$\frac{hc}{\lambda'} = 13.6Z^2\left(\frac{1}{9} - \frac{1}{25}\right)$$

$$\lambda' = k \times \frac{9 \times 25}{16} \quad ; \quad \lambda' = \frac{3}{4}\lambda \times \frac{9 \times 25}{16} \quad = \frac{3^3 \times 5^2}{2^6}\lambda$$

4.(D) $\mu = \frac{A_m}{A_c} = 0.5 \qquad \{A_m = 20 \text{ volt}, \; A_c = 40 \text{ volt}\}$

$m(t) = A_m \sin \omega_m t \qquad \{\omega_c = 2\pi \times 10^6\}$

$c(t) = A_c \sin \omega_c t \qquad \{\omega_m = 2\pi \times 10 \times 10^3\}$

$C_m(t) = (A_c + A_m \sin \omega_m t)\sin \omega_c t \quad \Rightarrow \quad A_c\{1 + \mu \sin \omega_m t\}\sin \omega_c t$

5.(C) $R_1 = \frac{\rho_1 L}{a^2}$; $\qquad R_2 = \frac{\rho_2 L}{3a^2}$

Both are in parallel $\quad \therefore \quad R_{eq} = \frac{R_1 R_2}{R_1 + R_2} = \frac{\rho_1 \rho_2 L}{a^2(3\rho_1 + \rho_2)}$

6.(A) $I = \frac{p}{4\pi r^2} \qquad \ldots$ (i)

and $\quad I = \frac{1}{2} \epsilon_0 E_0^2 c \qquad \ldots$ (ii)

From (i) and (ii)

$$\frac{p}{4\pi r^2} = \frac{1}{2}\epsilon_0 E_0^2 c \Rightarrow \quad E_0 = \sqrt{\frac{2p}{4\pi \epsilon_0 r^2 c}}; \qquad E_0 = \sqrt{\frac{9 \times 10^9 \times 2 \times 15}{15 \times 15 \times 10^6 \times 3 \times 10^8}} = 2 \times 10^{-3} \text{ V/m}$$

7.(C) $v = \sqrt{\frac{\gamma RT}{M}}$; $\quad \frac{V_1}{V_2} = \sqrt{\frac{T_1}{T_2}}$; $\quad V_1 = \sqrt{\frac{363}{300}}(340) = (1.1)(340) = 374$ m/s

8.(D) Fact. Due to decrease in intermolecular forces

9.(B) Time taken by trains to collide $= \frac{300}{150} = 2$hrs

Speed of bird = 200 km/hr

Total distance travelled by bird $= 200 \times 2 = 400$ km

10.(B) At minima, $I = (\sqrt{I_0} - \sqrt{25I_0})^2 = 16I_0$

At maxima, $I = (\sqrt{I_0} + \sqrt{25I_0})^2 = 36I_0 \quad \therefore \qquad \text{Ratio} = \frac{16}{36} = \frac{4}{9}$

11.(C) The time spent in magnetic field (t) is independent of velocity of charge.

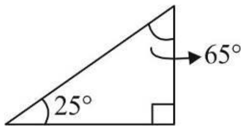
$$t = \frac{1}{4}\left(\frac{2\pi m}{qB}\right)$$

12.(C) At second surface,

$i = 25°$

$r = 25°$

$\therefore \qquad \theta = i + r = 50°$



13.(A) Workdone by gas = area enclosed in P-V diagram

$= \pi(2)^2$ lt-atm

14.(C) Taking torque at top pt, $\quad mg\sin 30° \times L/2 = F\cos 30° \times L \Rightarrow F = mg/2\sqrt{3}$

15.(B) For permanent magnetic Y is good choice, As it has very high coercivity (comparitively)

16.(C) Let the origin be at the CM of the particles, let the initial positions of the particles be $x = \frac{a}{2}$ and $x = -\frac{a}{2}$ and let the instantaneous positions of the particles be $x = r$ and $x = -r$

Let the instantaneous velocity of each particle be v

Let the time after which the distance between the particles has reduced to $\frac{a}{2}$ be T

Then, for the particle that was initially at $x = \frac{a}{2}$,

$$\frac{Gm^2}{(2r)^2} = -m\frac{v\,dv}{dr} \qquad \Rightarrow \qquad \frac{Gm}{r^2} = -\frac{4v\,dv}{dr}$$

$$\Rightarrow \qquad -4\int_0^v v dv = Gm\int_{a/2}^{r}\frac{dr}{r^2} \qquad \Rightarrow \qquad v^2 = \frac{Gm}{2}\left(\frac{1}{r} - \frac{2}{a}\right) \qquad \Rightarrow \qquad v = -\left[\frac{Gm}{2}\left(\frac{1}{r} - \frac{2}{a}\right)\right]^{1/2}$$

[v is negative because the velocity is towards the $-X$ direction]

$$\Rightarrow \qquad \frac{dr}{dt} = -\left[\frac{Gm}{2}\left(\frac{1}{r} - \frac{2}{a}\right)\right]^{1/2} \Rightarrow -\int_{a/2}^{a/4}\sqrt{\frac{r}{a-2r}}\,dr = \sqrt{\frac{Gm}{2a}}\int_0^T dt$$

$$\Rightarrow \qquad \int_{a/2}^{a/4}\sqrt{\frac{r}{a-2r}}\,dr = -\sqrt{\frac{Gm}{2a}}\,T \qquad \ldots \text{(i)}$$

Let us now evaluate the integral $I = \int\sqrt{\frac{r}{a-2r}}\,dr$

Let $r = \frac{a\sin^2\theta}{2} \qquad \Rightarrow \qquad dr = a\sin\theta\cos\theta\, d\theta$

Therefore, $\qquad I = \frac{a}{\sqrt{2}}\int \sin^2\theta\, d\theta = \frac{a}{2\sqrt{2}}\int(1-\cos 2\theta)d\theta = \frac{a}{2\sqrt{2}}\left[\theta - \frac{1}{2}\sin 2\theta\right]$

Since $r = \frac{a\sin^2\theta}{2}$, $\theta = \sin^{-1}\sqrt{\frac{2r}{a}}$ and $\quad \sin 2\theta = 2\sin\theta\cos\theta = 2\left(\sqrt{\frac{2r}{a}}\right)\left(\sqrt{1-\frac{2r}{a}}\right) = \frac{\sqrt{8r(a-2r)}}{a}$

So, $I=\frac{a}{2\sqrt{2}}\left(\sin^{-1}\sqrt{\frac{2r}{a}}-\frac{\sqrt{2r(a-2r)}}{a}\right)=\frac{a}{2\sqrt{2}}\sin^{-1}\sqrt{\frac{2r}{a}}-\frac{\sqrt{r(a-2r)}}{2}$

Therefore, from equation (i),

$$T=-\sqrt{\frac{2a}{Gm}}\left(\frac{a}{2\sqrt{2}}\sin^{-1}\sqrt{\frac{2r}{a}}-\frac{\sqrt{r(a-2r)}}{2}\right)_{a/2}^{a/4}=-\sqrt{\frac{2a}{Gm}}\left(-\frac{a}{2\sqrt{2}}\left(\frac{\pi}{4}\right)-\frac{1}{2}\left(\frac{a}{2\sqrt{2}}\right)\right)$$

Hence, $T=\sqrt{\frac{a}{Gm}}\left(\frac{a\pi}{8}+\frac{a}{4}\right)=\left(\frac{\pi+2}{8}\right)\sqrt{\frac{a^3}{Gm}}$

17.(B) In y axis $u_y=v_0$ $a_y=-E_0q/m$

$$s_y=0,\ u_yt+\frac{1}{2}a_yt^2=0 \qquad \Rightarrow\ t=2u_y/a_y=\frac{2v_0m}{E_0q}$$

x coordinate at that time $=v_x\times t\ =\frac{2v_0m}{E_0q}\times v_0\ =\frac{2v_0^2m}{E_0q}$

18.(D) Pitch $p=0.5$ mm

Device has +ve zero error

$$R=6p+28\frac{p}{50}-4\frac{p}{50}\ =6p+\frac{24p}{50};\quad 3\text{mm}+\frac{24}{50}\times0.5\ =3.24\text{ mm}$$

19.(A) Centripetal acceleration is provided by friction force. Maximum friction force should be equal to or greater that mv^2/R.

$$\mu mg=\frac{mv^2}{R};\quad \mu=\frac{100}{100\times10};\quad \mu=1/10;\qquad \mu=0.1$$

20.(C) Using parallel axis theorem for the side BC, the moment of inertia of the triangle about an axis passing through A and perpendicular to the plane of the triangle,

$$I_A=2\left(\frac{1}{3}ma^2\right)+\left(\frac{1}{12}ma^2+m\left(\frac{\sqrt{3}a}{2}\right)^2\right)=\frac{3}{2}ma^2$$

If the angular velocity of the triangle at any instant is ω, the velocity of the vertex B at that instant is $a\omega$

Therefore, the velocity of B is maximum at the instant the angular velocity is maximum, i.e. when the side BC becomes horizontal

Let the angular velocity at this instant be ω_{MAX}

Then, by conservation of energy

Gain in kinetic energy = Loss in potential energy

$$\frac{1}{2}I_A\ \omega_{MAX}^2=3mg(\text{Loss in height of CM of triangle})$$

$$\frac{1}{2}\left(\frac{3}{2}ma^2\right)\omega_{MAX}^2=3mg\left(\frac{a}{\sqrt{3}}-\frac{a}{2\sqrt{3}}\right)=\frac{\sqrt{3}}{2}mga \qquad \Rightarrow \qquad \omega_{MAX}^2=\frac{2g}{\sqrt{3}a}$$

Therefore, the maximum velocity of the vertex B is given by $v_{MAX}^2=\omega_{MAX}^2a^2=\frac{2ga}{\sqrt{3}}$

SECTION-2

21.(7) In case 1, $E_{\text{photon}} = 2eV$; $\phi_{\text{metal surface}} = E_{\text{photon}} - KE_{\max} = 1eV$

In case 2, $E_{\text{photon}} = \dfrac{12400}{1550} = 8eV$; $\max.KE = 8eV - 1eV = 7eV$

22.(3.33) Charge initially on capacitor $1 = 10\mu C$

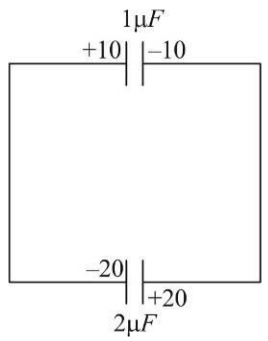
Charge initially on capacitor $2 = 20\mu C$

Finally the potential difference across both capacitor will be equal.

Let us assume that to be *V*.

$1V + 2V = +10$

$V = 10/3 = 3.33V$



23.(70) When relative sliding stops, both move with same velocity.
Only friction acts in horizontal direction.

$\therefore$ for system no external force, we can apply conservation of momentum.

$1 \times 12 + 5 \times 0 = 6 \times v$

$v = 2$ m/s

Now work done by friction on block = change in kinetic energy of block

$$w = \frac{1}{2}m(2^2 - 12^2) = \frac{1}{2}(1)(4-144) = \frac{1}{2}(-140) = -70J$$

24.(1.60) $x = \dfrac{8R}{2}$; $y = \dfrac{5R}{2}$; $\dfrac{x}{y} = \dfrac{8}{5} = 1.6$

25.(8) EMF induced $= BLV = 4 \times 2 \times 1 = 8V$

Current produced $= \dfrac{BLV}{R} = \dfrac{8}{1} = 8A$

26.(175) $t_{1/2}$, half life $= \dfrac{\log_e 2}{\lambda} = 3$ s

Initially 200 nuclei were there
After 3 half-lives, 9 seconds,

No. of nuclei $= \dfrac{1}{2^3}(200) = 25$

Therefore, the number of nuclei that decayed = 200 – 25 = 175

27.(5) Gold band is equal to 5% tolerance

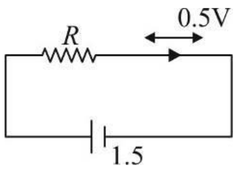
28.(21) $f_0 = 300\,\text{Hz}$

3^{rd} overtone $= 7f_0 = 2100$ Hz

29.(800) $\dfrac{Q_H}{T_H} = \dfrac{Q_L}{T_L}$ $Q_L = \dfrac{300}{600} \times 1600 = 800$ J

$\therefore$ work done $= Q_H - Q_L = 1600 - 800 = 800$ J

30.(100)



$1R + 0.5 = 1.5$

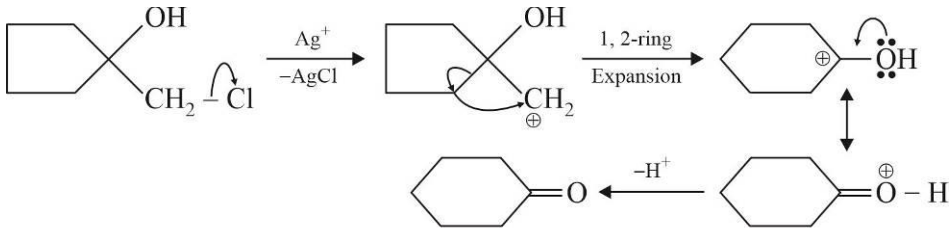
$1R = 1$

$R = \dfrac{1000}{10} = 100\Omega$

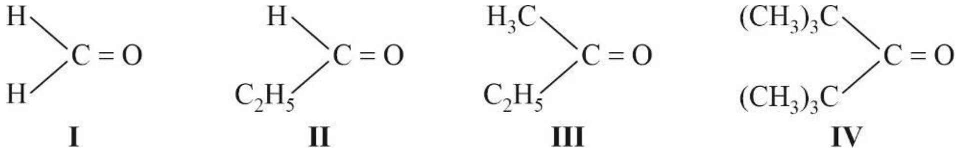
Chemistry

SECTION-1

1.(B) Here cyclohexanone is formed as follows



2.(D) If the carbonyl compound is sterically crowded, then it will be reluctant to undergo addition reaction. Moreover, attachment of bulkier alkyl group with the carbonyl carbon decreases the partial positive charge resulting into the minimization of attack by R^- from RMgBr. So, the order is



3.(B) $\underset{(P)}{CH_3CH_2COOH} + NH_3 \longrightarrow \underset{(Q)}{CH_3CH_2C\overset{-}{O}_2\overset{+}{N}H_4} \xrightarrow[-H_2O]{\Delta} \underset{(R)}{CH_3CH_2CONH_2}$

$\xrightarrow{KOH+Br_2} CH_3CH_2NH_2 + H_2O$

4.(A) **(A)** SiO_2 has three-dimensional network structure of $Si-O$ bonds; while carbon dioxide consists of discrete CO_2 molecules. SiO_2 is solid, whereas CO_2 is a gas

(B) Because of its great affinity for oxygen, Si always occurs as the oxide, silica (SiO_2) or in the form of silicates, which are the compounds of SiO_2 with other metal oxides.

(C) $SiO_2 + 2Mg \longrightarrow Si + 2MgO$

(D) The reluctance of silicon to form $p\pi - p\pi$ bonds to itself is clearly shown by the fact that silicon does not exist in graphite-like structure, but only in diamond like structure.

5.(D) $\lambda = \frac{h}{p}$, there is inverse relation between de-Broglie wavelength and momentum, hence the graph will be rectangular hyperbola.

6.(D) To stop bleeding, $FeCl_3$ is applied locally because $FeCl_3$ causes denaturation of proteins present in blood. It is a case of coagulation.

7.(A) **i.** $Zn + 2NaOH \longrightarrow Na_2ZnO_2 + H_2$

ii. $4Au + 8NaCN + O_2 + 2H_2O \longrightarrow 4Na[Au(CN)_2] + 4NaOH$

iii. $Cu + \underset{(conc.)}{4HNO_3} \longrightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$

iv. Formation of passive layer of Fe_2O_3 on the surface of Fe and NO_2 gas is evolved

8.(D) $[Co(NH_3)_6]^{3+}$ is diamagnetic $(t_{2g}^6 eg^0)$ and is inner orbital complex.

9.(A) When Ferric chloride reacts with potassium thiocyanate a blood red colour of Ferric thiocyanate is formed

$$FeCl_3 + 3KSCN \longrightarrow Fe(SCN)_3 + 3KCl$$

10.(D) As $Zn(OH)_2$, BeO, Al_2O_3 are amphoteric, so they can react with both HCl and NaOH.

11.(D) In S_N2 reaction, inversion takes place, so configuration get reversed.

12.(A)

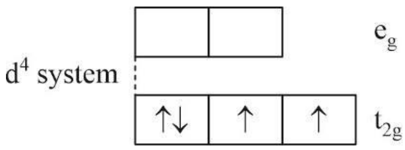
$$\overset{5}{C}H_3-\overset{4}{C}=\overset{3}{C}H-\overset{2}{C}H_2-\overset{1}{C}OOH$$
$$\quad\;|$$
$$C-O-C_2H_5$$
$$\|$$
$$O$$

4-ethoxycarbonylpent-3-enoic acid

13.(D) The given value of μ (spin only)

$$2.84 = \sqrt{n(n+2)}\,BM,\; So,\; n = 2$$

Among the given configurations, d^4 system in strong field ligand will have 2-unpaired e^- in t_{2g} set of orbitals as shown below.



14.(C) The process of moving sodium and potassium ions across the cell membrane is an active transport process involving the hydrolysis of ATP to provide the necessary energy.

15.(D) Nitrogen, sulphur and halogens are tested in an organic compound by lassaigne's test.
The organic compound is fused with sodium metal as to convert these elements into ionisable inorganic substances.

$$Na + C + N \longrightarrow NaCN$$

$$2Na + S \longrightarrow Na_2S$$

$$2Na + X_2 \longrightarrow 2NaX$$

The cyanide, sulphide or halide ions can be confirmed in aqueous solution by usual test.

16.(A) The second ionization energy of K is maximum, because the second electron is removed from fully filled $3p^6$ subshell. The second ionization energies of group 2 elements decrease down the group. Hence, second I.E. of Ca $>$ second I.E. of Ba.

17.(A) All monosaccharides which differ in configuration at C_1 and C_2 gives the same osazone. Since, glucose and fructose differ from each other only in configuration at C_1 and C_2 therefore, they give the same osazone. All other options given in the questions do not satisfy this condition and hence, do not from the same osazone.

18.(B) $2CO(g) + O_2(g) \longrightarrow 2CO_2(g)$

$\Delta n_g = 2-(2+1) = -1$

$\Delta H = \Delta E + \Delta n_g RT$ or $\Delta H = \Delta E - 1RT$

i.e. $\quad \Delta H < \Delta E$

19.(D) Sometimes, when maximum covalency is obtained, the halides become inert to water, thus SF_6 (or similarly CCl_4) is stable. This is because SF_6 is coordinately saturated and sterically hindered. Thus, SF_6 is inert to water, because of kinetic rather than thermodynamic factor.

A. $PCl_5 + 4H_2O \longrightarrow H_3PO_4 + 5HCl$

B. $SiCl_4 + 4H_2O \longrightarrow Si(OH)_4 + 4HCl$

C. $BCl_3 + 3H_2O \longrightarrow B(OH)_3 + 3HCl$

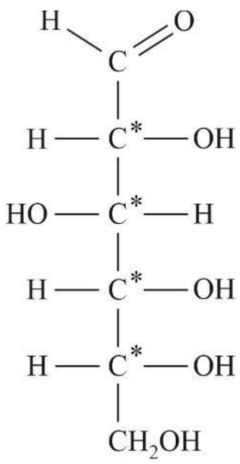
20.(B) One mole of water is converted to vapour at its boiling point which is 100°C and at 1 atm. For this process $\Delta G = 0.$ As phase transformation of water is an equilibrium process and at equilibrium, free energy change is always zero.

SECTION-2

21.(5) ATP has phosphate group, PO_4^{3-}

So $x + 4(-2) = -3$ or $x = +5$

22.(16)



The number of chiral carbons in open-chain aldohexose (such as glucose) is four, therefore, the number of stereoisomers $= 2^4 = 16.$

23.(–836)

$$\Delta H_{rxn} = \Delta H_f^\circ (N_2O, g) + 3\Delta H_f^\circ (CO_2, g) - (2\Delta H_f^\circ NO_2, g) - 3\Delta H_f^\circ (CO, g)$$
$$= 81 + 3 \times (-393) - 2 \times 34 - 3(-110)$$
$$= 81 - 1179 - 68 + 330 = -836\,kJ$$

24.(0) As 3f-orbital does not exist as value of ℓ cannot be equal to n, hence no electron can be accommodated.

25.(400) As 500g of tooth-paste has $= 0.2\,g$ of F^-

So, $10^6\,g$ of toothpaste has $= \dfrac{0.2\,g \times 10^6\,g}{500\,g} F^- = 400\,g$ of F^-

Hence, concentration in ppm is 400.

26.(150) $P_1 = 1\text{atm}$

$$P_2 = 1\text{atm} \times \frac{40}{100} = 0.40\,\text{atm}$$

$V_1 = 100\,\text{cm}^3$

$V_2 = ?$

At constant temperature, $P_1V_1 = P_2V_2$

So $V_2 = \dfrac{P_1V_1}{P_2} = \dfrac{1\text{atm} \times 100\,\text{cm}^3}{0.40\,\text{atm}} = 250\,\text{cm}^3$

Hence, the volume of bulb $B = (250 - 100)\,\text{cm}^3 = 150\,\text{cm}^3$

27.(4145.40)

$$\Delta G^\circ = -2.303RT \log K_p = -2.303 \times 2 \times 300 \log 10^{-3}$$

$$= -2.303 \times 2 \times 300 \times (-3) = 4145.4\,\text{cal mol}^{-1}$$

28.(1.93)

To decompose 1 mole of H_2O, we need two moles of electrons i.e. 2 faraday

To decompose 4 moles of H_2O, we need 8 faraday.

Now, $Q = i \times t \quad \Rightarrow \quad t = \dfrac{8 \times 96500}{4} = 1.93 \times 10^5\,\text{sec}$

29.(0.50) Number of mol of A reacted $= 1 \times 10^{-5} = 10^{-5}$

Number of molecule of $A = 10^{-5} \times 6 \times 10^{23}$

Quantum efficiency $= \dfrac{6 \times 10^{18}}{1.2 \times 10^{19}} = 0.50$

30.(64) Temperature coefficient is 2 that means rate of reaction doubles at every 10°C rise in temperature.

Thus, $\dfrac{k_{90^\circ C}}{k_{30^\circ C}} = 2^6 = 64$

Mathematics

SECTION-1

1.(C) $p(10hr) = .1 \qquad p(s/10hr) = .8$

$p(7hr) = .2 \qquad p(s/7hr) = .6$

$p(4hr) = .7 \qquad p(s/4hr) = .4$

$p(s) = .1\times.8 + .2\times.6 + .7\times.4$

$$p(4hr/s) = \frac{.7\times.4}{.1\times.8+.2\times.6+.7\times.4} = \frac{28}{8+12+28} = \frac{28}{48} = \frac{7}{12}$$

2.(B) $\Delta \neq 0 \qquad \begin{vmatrix} 1 & -\lambda & -1 \\ \lambda & -1 & -1 \\ 1 & 1 & -1 \end{vmatrix} \neq 0$

$1(1+1) + \lambda(-\lambda+1) - 1(\lambda+1) \neq 0$

$2 - \lambda^2 + \lambda - \lambda - 1 \neq 0$

$\lambda^2 = 1 \Rightarrow \lambda = 1, -1$

$a^2 + b^2 = 2$

3.(C) b_1, b_2, b_3

$a, ar, ar^2........$

$a + ar = 1 \Rightarrow a = \dfrac{1}{(1+r)}$

$\sum_{k=1}^{\infty} b_k = 2 \Rightarrow \dfrac{a}{1-r} = 2$

$\dfrac{1}{1-r^2} = 2$

$r^2 = \dfrac{1}{2} \Rightarrow r = \pm\dfrac{1}{\sqrt{2}}$

$b_2 < 0 \Rightarrow r = -\dfrac{1}{\sqrt{2}} \quad \Rightarrow a = b_1 = 2 + \sqrt{2}$

4.(D) $(I+M)^2 = (I+M)(I+M) = I + 2M + M^2 = (I+2M)$

$(I+M)^3 = (I+2M)(I+M) = I + 3M + 2M^2 = (I+3M)$

$(I+M)^{50} = I + 50M$

$\det((I+M)^{50} - 50M) = \det(I) = 1$

5.(C) $y^2 = 4x \qquad (x-3)^2 + y^2 = 9$

$y = mx + \dfrac{1}{m} \qquad (3, 0), \ r = 3$

$$\left|\frac{3m + \frac{1}{m}}{\sqrt{1+m^2}}\right| = 3$$

$9m^2 + \frac{1}{m^2} + 6 = 9 + 9m^2$

$\frac{1}{m^2} = 3 \Rightarrow m = \pm\frac{1}{\sqrt{3}}$

$m = \frac{1}{\sqrt{3}}$ in first quadrant

$y = \frac{x}{\sqrt{3}} + \sqrt{3} \Rightarrow \sqrt{3}y = x + 3$

6.(B) $\frac{T_{r+1}}{T_r} = \frac{1}{3}$

$\Rightarrow \quad T_1, T_2, T_3$ ………..G.P.

$T_7 = \frac{1}{243}$

$ar^6 = \frac{1}{243}$

$\frac{a}{3^6} = \frac{1}{3^5} \Rightarrow a = 3$

Sum of infinite series $= T_1T_2 + T_2T_3 + T_3T_4$ …………..

$\frac{T_1T_2}{1-r^2} = \frac{3\times 3\frac{1}{3}}{1-\frac{1}{9}} = \frac{27}{8}$

7.(A) $\sim(p \vee (\sim p \vee q)$

$=\sim p \wedge (\sim(\sim p \vee q)) = \sim p \wedge (p \wedge \sim q)$

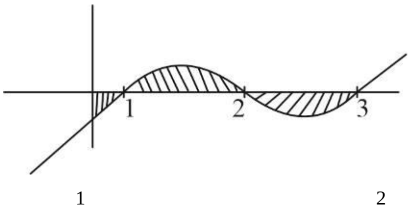
8.(B) $y = \frac{3}{2}\sin 2\theta$

$x = e^{\theta}\sin\theta$

$\frac{dy}{dx} = \frac{3\cos 2\theta}{e^{\theta}(\cos\theta + \sin\theta)} = \frac{3}{e^{\theta}}(\cos\theta - \sin\theta)$

9.(C) $(v_{ar})_{New} = k^2(v_{ar})_{Old} \quad = 4\times 5 = 20$

10.(B) $\int_0^3 |(x-1)(x-2)(x-3)|dx = \int_0^3 |x^3 - 6x^2 + 11x - 6|dx$



$A = -\int_0^1 (x^3 - 6x^2 + 11x - 6)dx + \int_1^2 (x^3 - 6x^2 + 11x - 6)dx - \int_2^3 (x^3 - 6x^2 + 11x - 6)dx$

$A = \frac{11}{4}$

11.(A) $|z-2|^2 = |z+2|^2$

$\Rightarrow \quad x = 0$

Hence $|z|$ minimum '0'

12.(C) $f(x) = \lambda(x-2)^2$

$\Rightarrow \quad 12 = \lambda(2)^2 \Rightarrow \lambda = 3$

$f(x) = 3(x-2)^2 \quad ; \quad f(6) = 3 \times 4^2 = 48$

13.(B) $\tan^{-1}(x) = t$

$t^2 - 4t + 3 > 0$

$t \in (-\infty, 1) \cup (3, \infty)$

$\tan^{-1}(x) \in \left(-\frac{\pi}{2}, 1\right)$

$x \in (-\infty, \tan 1)$

Largest integral $x = 1$

14.(A) $\log x = t$

$\left(t^2 - \frac{1}{t}\right)^{12}$

$T_{r+1} = {}^{12}C_r (t^2)^{12-r} \left(-\frac{1}{t}\right)^r = {}^{12}C_r\, t^{24-3r} (-1)^r$

For constant term $r = 8$ & coefficient ${}^{12}C_8$

$$\frac{12 \times 11 \times 10 \times 9}{24} = 45 \times 11 = 495$$

15.(B) $\frac{dy}{\sqrt{1-y^2}} = \frac{dx}{x^2}$

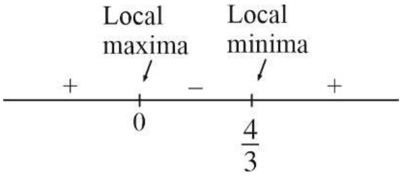
$\sin^{-1}(y) = -\frac{1}{x} + c \Rightarrow c = \frac{\pi}{2}$

$\sin^{-1}(y) = -\frac{\pi}{3} + \frac{\pi}{2} = \frac{\pi}{6}$

$y = \frac{1}{2}$

16.(C) $f(x) = \frac{x^5}{20} - \frac{x^4}{12} + 5$

$f'(x) = \frac{x^4}{4} - \frac{x^3}{3} = x^3\left(\frac{x}{4} - \frac{1}{3}\right)$



$f''(x) = x^3 - x^2 = x^2(x-1)$

$x = 1$ point of inflection

17.(C) $f'(x) = x(1+y)$

$\frac{dy}{y+1} = xdx$

$\ln(y+1) = \frac{x^2}{2} + c$

(0, 0)

$c = 0$

$y+1 = e^{\frac{x^2}{2}} \Rightarrow e^{\frac{x^2}{2}} = 2$

$\frac{x^2}{2} = k = (\ln 2 > 0)$

$x = \pm\sqrt{2k}$

18.(D) For point of intersection

$1 + 3\lambda = 3 + \mu$

$2 + \lambda = 1 + 2\mu$

$5\lambda = 5 \Rightarrow \lambda = 1, \mu = 1$

Point of intersection (4, 3, 5)

For greatest distance from origin perpendicular from meet plane at point of intersection

Hence equation $\vec{r} \cdot (4\hat{i} + 3\hat{j} + 5\hat{k}) = 50$

19.(D) Differentiable $\Rightarrow$ continuous

Continuous at $x = 2 \Rightarrow 2a + b = 0$

Continuous at $x = 3$

$0 = 9p + 3q + 1$

Differentiable at $x = 2$

$a = 2 \times 2 - 5 \Rightarrow a = -1$

Differentiable at $x = 3$

$2 \times 3 - 5 = 2p \times 3 + q \Rightarrow 6p + q = 1$

$a = -1, b = 2, p = \frac{4}{9}, q = -\frac{5}{3}$

20.(C) $2a^2 + 3b^2 = 35$

$a = 0, b \neq I$

$a = \pm 1, b^2 = 11 \Rightarrow b \neq I$

$a = \pm 2, b \ \ = 9 \Rightarrow b = 3 \text{ or } -3$

$a = \pm 3, b^2 = \frac{17}{3} \Rightarrow b \neq I$

$a = \pm 4, b^2 = 1 \Rightarrow b = \pm 1$

SECTION-2

21.(570) All even + 2 odd 1 even

${}^{10}C_3 + {}^{10}C_2 \times {}^{10}C_1$

$\frac{10 \times 9 \times 8}{6} + \frac{10 \times 9}{2} \times 10$

$120 + 450 = 570$

22.(12) Family of circle through (2, 2) and (9, 9), $(x-2)(x-9)+(y-2)(y-9)+\lambda(y-x)=0$

Touches $y=0$ (x-axis)

$(x-2)(x-9)+18-\lambda x=0$

$x^2-11x-\lambda x+36=0$

$x^2-(11+\lambda)x+36=0$ has repeated roots

$D=0 \Rightarrow \lambda=1, \lambda=-23$

$x=6$ or $x=-6$

Difference = 12

23.(2) $\left|3\vec{a}+4\vec{b}\right|^2=9|\vec{a}|^2+16\left|\vec{b}\right|^2+24\vec{a}\cdot\vec{b}$

But $\vec{a}\cdot\vec{b}=0,\quad |\vec{a}|=\left|\vec{b}\right|=k$

$\left|3\vec{a}+4\vec{b}\right|=5k$

$\left|4\vec{a}-3\vec{b}\right|=5k$

$10k=20 \Rightarrow k=2=|\vec{a}|=\left|\vec{b}\right|$

24.(8) $\int_0^{\pi/2}\frac{x}{2}|\cos 2x|\,dx$

$\int_0^{\pi/4}\frac{x}{2}\cos 2x\,dx-\int_{\pi/4}^{\pi/2}\frac{x}{2}\cos 2x\,dx$

$\left.\frac{x}{2}\frac{\sin 2x}{2}\right]_0^{\pi/4}-\int_0^{\pi/4}\frac{\sin 2x}{4}dx-\left[\frac{x}{2}\frac{\sin 2x}{2}\right]_{\pi/4}^{\pi/2}+\int_{\pi/4}^{\pi/2}\frac{\sin 2x}{4}dx$

$\frac{\pi}{16}+\left.\frac{\cos 2x}{8}\right]_0^{\pi/4}+\frac{\pi}{16}-\left[\frac{\cos 2x}{8}\right]_{\pi/4}^{\pi/2}$

$\frac{\pi}{8}-\frac{1}{8}-\left[-\frac{1}{8}\right]=\frac{\pi}{8}$

25.(4) $\lim_{x\to 0}\frac{ae^{2x}-b\cos x+c}{\frac{x^2\sin x}{x}}=1$

$\lim_{x\to 0}\frac{ae^{2x}-b\cos x+c}{x^2}\lim_{x\to 0}\frac{1}{\left(\frac{\sin x}{x}\right)}$

For limit to exist

$a=0, b=c$

$\frac{c}{2}=1 \Rightarrow c=2, b=2$

$a+b+c=4$

26.(220) ${}^4C_r+{}^4C_{r+1}={}^5C_{r+1}$

${}^4C_{r+1}={}^5C_{r+1}-{}^4C_r$

$$\sum_{r=0}^{3} {}^8C_r \cdot {}^4C_{r+1} = {}^8C_0 \cdot {}^4C_1 + {}^8C_1 \cdot {}^4C_2 + {}^8C_2 \cdot {}^4C_3 + {}^8C_3 \cdot {}^4C_4$$

$$= {}^8C_8 \cdot {}^4C_1 + {}^8C_7 \cdot {}^4C_2 + {}^8C_6 \cdot {}^4C_3 + {}^8C_5 \cdot {}^4C_4$$

$$= {}^{12}C_9 = \frac{12 \times 11 \times 10}{6} = 220$$

27.(0) $A^2 - 4A + 4I = 0$

$A^2 - 2AI - 2A + 4I = 0$

$(A - 2I)^2 = 0 \quad \Rightarrow \ |A - 2I| = 0$

$B = 297A - 594I$

$= 297(A - 2I)$

28.(16) $1 + x^4 - x^5 = a_0(1+x)^0 + a_1(1+x) + a_2(1+x)^2 .... + a_5(1+x)^5$

Differentiate

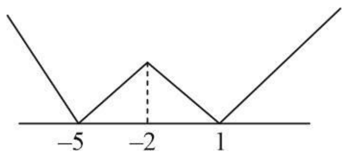
$$4x^3 - 5x^4 = a_1 + 2a_2(1+x) + 3a_3(1+x)^2 \ldots\ldots\ldots\ldots$$

$12x^2 - 20x^3 = 2a_2 + 6a_3(1+x) \ldots\ldots\ldots\ldots$

Put $x = -1$

$12 + 20 = 2a_2 .... \Rightarrow a_2 = 16$

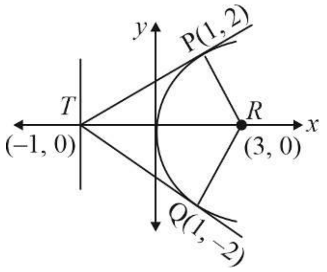
29.(9)



$c = -5,\ a = 2,\ b = 1$

30.(8) PQ is focal chord.

Quadrilateral PTQR is square



$$\text{Area} = \frac{PQ \times TR}{2} = 8$$